



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Roll Number	2520030550
Branch	CSE
Course Outcome	CO4

Case Study Topic: Waste Sense – Smart Waste Collection & Recycling Optimization System Using Dijkstra's

Work Done Summary:

The Waste Sense system uses Dijkstra's Algorithm to determine the shortest and most cost-effective routes for waste collection vehicles. The algorithm helps reduce transportation costs, fuel consumption, and collection time by identifying the minimum-distance path between collection centers, smart bins, and recycling plants.

Implementation Details:

1. Create a weighted graph for the waste collection network.
2. Add smart bins, collection centres and recycling plants as nodes.
3. Add roads with transportation costs as weighted edges.
4. Select a source collection centre.
5. Apply Dijkstra's Algorithm to calculate the shortest distance to all locations.
6. Display the optimal waste collection routes and their costs.

Result: Screenshot/output:

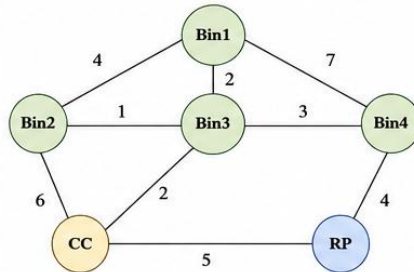
Student Signature	Faculty Signature

Shortest distances from CC:

Node	Distance(km)	Path
CC	0	CC
Bin1	4	CC -> Bin3 -> Bin1
Bin2	3	CC -> Bin3 -> Bin2
Bin3	2	CC -> Bin3
Bin4	5	CC -> Bin3 -> Bin4
RP	5	CC -> RP

Example: Waste Collection Network (Weighted Graph)

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in4), 1 collection center (CC),
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ce to all locations.
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Edges and Weights (Distance in km)			
Edge	From	To	Distance (km)
E1	Bin1	Bin2	4
E2	Bin1	Bin3	2
E3	Bin1	Bin4	7
E4	Bin2	Bin3	1
E5	Bin3	Bin4	3
E6	Bin2	CC	6
E7	Bin3	CC	2
E8	Bin4	RP	4
E9	CC	RP	5

Goal Example:

Find the shortest path and minimum cost from Collection Center (CC) to all other locations (Bin1, Bin2, Bin3, Bin4, RP) using Dijkstra's Algorithm.

Dijkstra's Algorithm Steps from Source (CC)

Step	Visited Set	Current Node	Distance Update
0	{CC}	CC	Init: dist(CC)=0, others = ∞
1	{CC, Bin3}	Bin3	Update: Bin3=2, Bin1=4, Bin2=6, Bin4=5, RP=5
2	{CC, Bin3, Bin2}	Bin3	Update: Bin2=3 (via Bin3)
3	{CC, Bin3, Bin2, Bin1}	Bin2	Update: Bin1=4 (no change)
4	{CC, Bin3, Bin2, Bin1, RP}	RP	Update: RP=5 (no change)
5	{CC, Bin3, Bin2, Bin1, RP, Bin4}	Bin4	Update: Bin4=6 (via Bin3)

Shortest Path from CC to RP:
CC -> RP with total cost = 5 km

Shortest Paths from CC to All Locations

Destination	Shortest Distance (km)	Shortest Path
CC	0	CC
Bin1	4	CC -> Bin2 -> Bin1
Bin2	3	CC -> Bin3 -> Bin2
Bin3	2	CC -> Bin3
Bin4	6	CC -> Bin3 -> Bin4
RP	5	CC -> RP

Explanation:

- The shortest route from CC to RP is CC -> RP with cost 5 km.
- The shortest route from CC to Bin1 is CC -> Bin2 -> Bin1 with cost 4 km.
- Dijkstra's algorithm ensures minimum cost routes to all locations.

- Dijkstra's Algorithm provides an effective solution for route optimization in smart waste management systems. By selecting the shortest paths, the Waste Sense system ensures efficient waste collection, reduced operational expenses, and better utilization of transportation resources.

GIT HUB LINK : <https://github.com/Mahaiswaryakaruturi/DSA2-Case-Studies>

Student Signature

Faculty Signature

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